Predicting 30-Day Hospital Readmission Risks

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Abstract

In this project, the Diabetes 130-US Hospital dataset, sourced from the UCI Machine Learning Repository, is explored to predict 30-day hospital readmission risks. Various machine learning models, including logistic regression, decision trees, random forests, gradient boosting, k-nearest neighbors, and neural networks, were used to analyze this binary classification problem. The dataset was highly imbalanced; therefore, the results were evaluated using the F1 score and ROC-AUC. Logistic Regression was tuned for hyperparameters and achieved the strongest performance with an F1 score of 0.2657 and ROC-AUC of 0.6570. A notable result was that some of the tree-based models achieved high ROC-AUC but very low F1 scores, indicating the complexities involved in an imbalanced dataset. Results pointed out that simpler and more interpretable models can outperform more complex models when working with complicated datasets. Overall, this machine learning system is best used as a decision support tool for clinicians, as when working in the medical industry, the ultimate decision factor should fall upon the clinician.

Introduction

Hospital readmissions are problematic for the healthcare industry as high levels of them can lead to higher costs, poorer patient outcomes, and reduced efficiency. However, the ability to know if a patient is at a higher likelihood of being readmitted can aid hospitals in improving discharge planning and resource utilization.

The objective of this project is to create a machine learning pipeline that can predict whether a patient will be readmitted within 30 days. This task is framed as a binary classification problem, where the positive case is readmissions within 30 days.

To tackle this, a machine learning pipeline that includes exploratory data analysis, preprocessing, model comparison, hyperparameter tuning, and interpretation is implemented. We evaluated the performance of multiple models, measuring the F1 score and ROC-AUC, which provide the best insights for highly imbalanced datasets.

Related Work

The Diabetes 130-US Hospitals dataset is a resource that has been used widely for assessing hospital readmission prediction. A lot of prior work has applied many of the same machine learning models used in this project, including logistic regression, random forests, gradient boosting, and neural networks.

Most prior studies have reported moderate predictive performance with ROC-AUC values around 0.65. In many of these studies, the more complex models, such as neural networks and ensemble methods, have only provided minor improvements in results; however, performance has always been an issue due to class imbalance and data quality.

Prior work tends to focus on a single model and attempts to make it as strong as possible. In this project, multiple model types are evaluated using metrics suited best for imbalanced classification. Overall, results are consistent with prior work, showing readmission prediction as a challenging problem and that simpler models perform very strongly.

Dataset and Problem Formulation

The Diabetes 130-US Hospitals dataset from the UCI Machine Learning Repository contains 101,766 patient encounters and 47 features collected from 130 U.S. hospitals from 1999 to 2008. Some of the features include demographic information, admission details,

diagnosis codes, medications, and prior hospital utilization.

On the original dataset, there is a target variable called "readmitted". It contains three categories: no readmission, readmission after 30 days, and readmission within 30 days. For this project, the problem was converted into a binary classification task, where readmission within 30 days was the positive class and labeled "1", while all other outcomes were the negative class and labeled "0".

One of the most important facts about this dataset is that it is highly imbalanced, with only about 11% of patient encounters belonging to the positive class, "readmitted within 30 days". This imbalance was a challenge throughout this project, and it influenced the modeling approach and evaluation metrics used.

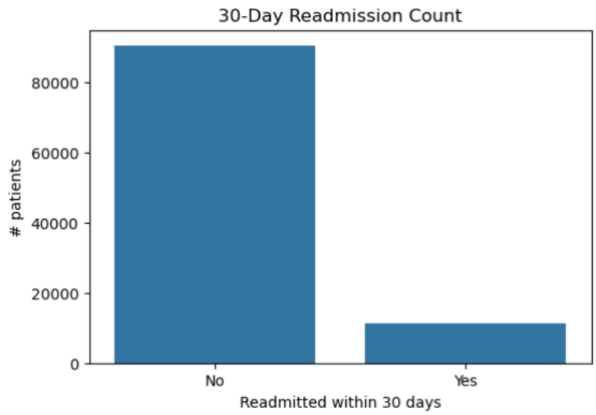

Figure 1: Distribution of 30-Day Hospital Readmission

The dataset is highly imbalanced, with only about 11% of patients readmitted within 30 days.

Methodology

This system follows a structured pipeline which allows for reproducibility and models to work with new data. The first step in the pipeline is exploratory data analysis, which was done to understand feature distributions, find missing values, and observe any patterns in the data. Next, data preprocessing included replacing placeholder null values, dropping negligible columns, and one-hot encoding categorical variables. Many of the missing values appeared as true missing values and as placeholder "?". As for the columns dropped, they included too many missing values to the point that they were unusable; some examples of these are A1Cresult, max_glu_serum, and payer_code. From here, to reduce dimensionality, secondary and tertiary diagnosis features were removed, leaving only the primary diagnosis column "diag_1", while removing "diag_2" and "diag_3".

After this, a stratified train/test split was used to preserve the class distribution, which was essential considering the high imbalance of the dataset. Doing this ensured that both training and testing sets contained approximately the same proportion of readmission cases.

Models including logistic regression, decision trees, random forests, gradient boosting, k-nearest neighbors, and neural networks were then trained. These models, in specific were chosen as they give a good representation of a range of different approaches which include linear models, tree-based ensemble methods, distance-based models, and neural networks. This allows for comparison between different model types and families.

Due to the overall class imbalance, the F1 score and ROC-AUC were the main evaluation metrics for how well the model was able to predict readmission cases. Accuracy was not used, as many models achieved high accuracies by predicting the majority class.

Lastly, using GridSearchCV on logistic regression, hyperparameter tuning was performed, which optimized for the F1 score. This was done on logistic regression as it showed the best preliminary results. This allowed for the best comparison between baseline and tuned performance.

Results and Evaluation

Overall, the results showed that logistic regression provided the best balance between performance across all models. The baseline logistic regression model achieved an F1 score of 0.256 and an ROC-AUC of 0.642. After hyperparameter tuning, there was a slight improvement for logistic regression with an F1 score of 0.267 and ROC-AUC of 0.659.

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.6454	0.1677	0.5495	0.2569	0.6428
Decision Tree	0.7914	0.1610	0.2065	0.1809	0.5357
Neural Network	0.8051	0.1482	0.1572	0.1526	0.5441
KNN	0.8788	0.2662	0.0489	0.0826	0.5705
Gradient Boosting	0.8885	0.5294	0.0079	0.0156	0.6776
Random Forest	0.8887	0.6667	0.0053	0.0105	0.6537

Table 1. Model Performance Comparison

This table summarizes performances across all models tested in this project. Logistic Regression provided the strongest F1 score.

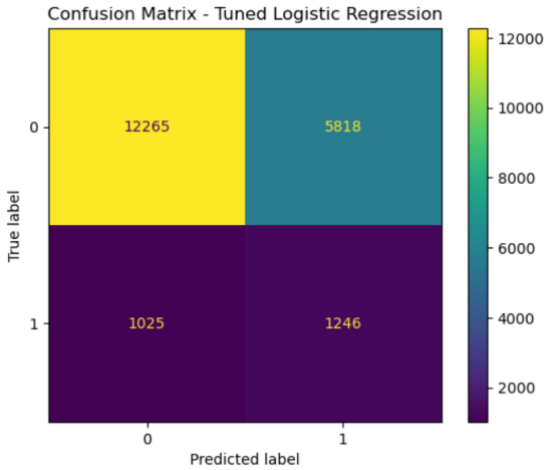

Figure 2. Confusion Matrix for Tuned Logistic Regression

This figure illustrates the classification performance for the tuned logistic regression model. While the model was able to correctly classify a large part of the majority class, it struggled to catch all positive readmission cases.

Some of the more complex models, such as random forest and gradient boosting, achieved higher accuracy but with very low F1 scores. This was most likely due to these models predicting from the majority class, thus resulting in the low recall for the minority class. This data makes sense; the dataset has only 11% observations belonging to the positive class.

One notable result was that gradient boosting achieved the highest ROC-AUC of 0.677 while having an extremely low F1 score of 0.015. These results mean that the gradient boosting model was able to rank observations but predicted only a few cases as positive. This is evident as the recall for the minority class was extremely low (0.007), thus causing the F1 score to drop.

EDA also pointed out that most patients had very few prior inpatient or emergency visits. This explains why models struggled to identify the minority class and defaulted to predicting the majority.

These findings highlight the importance of using the correct evaluation metrics. If accuracy were used as the core metric for an imbalanced dataset such as this one, the results would be incorrect. It would not correctly reflect the model's true ability to detect the positive class. These results are consistent with prior work using this dataset. Most reports mention moderate predictive performance, thus reinforcing the fact that hospital readmission predictions are a very challenging problem to tackle.

Model Interpretation

To get a better understanding of the model's predictions, the coefficients of the logistic regression model were analyzed to find the strongest positive and negative predictors of readmission. For the strongest positive predictors of readmission, prior inpatient visits, certain types of diagnosis categories, and some medical specialties were strongly associated. This makes sense as certain diagnosis categories, such as something cancer-related, often require strict supervision and lead to readmission. These positive predictor features suggest that the model was able to

capture patterns for patients and their prior healthcare needs.

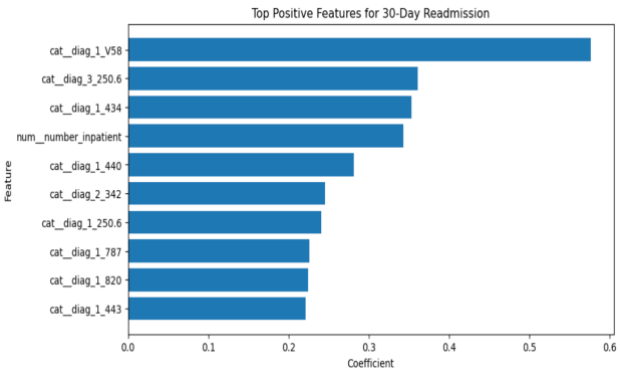

Figure 3. Top Positive Features for 30-Day Readmission

This figure shows the most influential positive features for 30-day readmission that were identified by logistic regression.

Earlier observations stated that patients with more prior inpatient visits were more likely to be readmitted, which aligns with the results seen from analyzing positive predictors. This further reinforces the importance of these utilization-based features for predicting readmission.

Some of the negative predictors included younger age groups and certain specialties such as orthopedics. These were associated with lower readmission risk as either the reason for hospitalization in the first place was not too serious or the patient was healthy.

Overall, the model relied heavily on utilization and diagnosis-related features. This is consistent with patterns that were observed during early EDA. However, in industries such as the medical one, these are best interpreted as associations rather than relationships.

Impact and Limitations

The impact of a machine learning model that can predict if a patient will be readmitted within 30 days is very useful for hospitals and clinicians, as they can better serve their patients and improve outcomes. The results of this project were moderate; however, these results still show that it can learn and predict the correct outcome. This model will likely need to improve before being used universally, but the results thus far are strong enough to provide many valuable insights for clinicians.

The limitations of this project stem from the dataset itself being imbalanced. Prediction is generally difficult in datasets such as this, and it can also bias models towards the majority class. Some of the features in the dataset, such as race, may cause the models to pick up on systematic patterns rather than actual clinical factors.

Due to these limitations, a machine learning model such as this in the medical industry should be used as a decision support tool for clinicians to better utilize their resources and plan accordingly for their patients.

Conclusion

In conclusion, this project explores a real-world healthcare dataset that predicts 30-day hospital readmission risks through a machine learning pipeline. This pipeline starts from preprocessing, model comparison, hyperparameter tuning, and then ends with interpretation. Several models were tested, including logistic regression, decision trees, neural networks, k-nearest neighbors, gradient boosting, and random forest. From these models, logistic regression had the strongest performance.

While the results were moderate, they were consistent with previous works. This reinforced that hospital readmission prediction is a challenging problem. This project also highlights several important predictors of readmission, such as diagnosis-related features, as well as prior inpatient utilization. For this imbalanced dataset, accuracy was a misleading metric, so the strongest performance was dependent on F1 score and ROC-AUC.

One key takeaway from this project is that model evaluation is just as important as model choice. Future work should focus on improving feature engineering, stronger imbalance handling methods, and threshold tuning to ultimately improve the performance and make models more useful and accurate as clinical decision support tools.

Project Code

The code and dataset for this project can be found below:

Code — https://github.com/anshcpatel11/hospital-readmissions-ml

Datasets — https://archive.ics.uci.edu/dataset/296/diabetes+130-us+hospitals+for+years+1999-2008

References

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